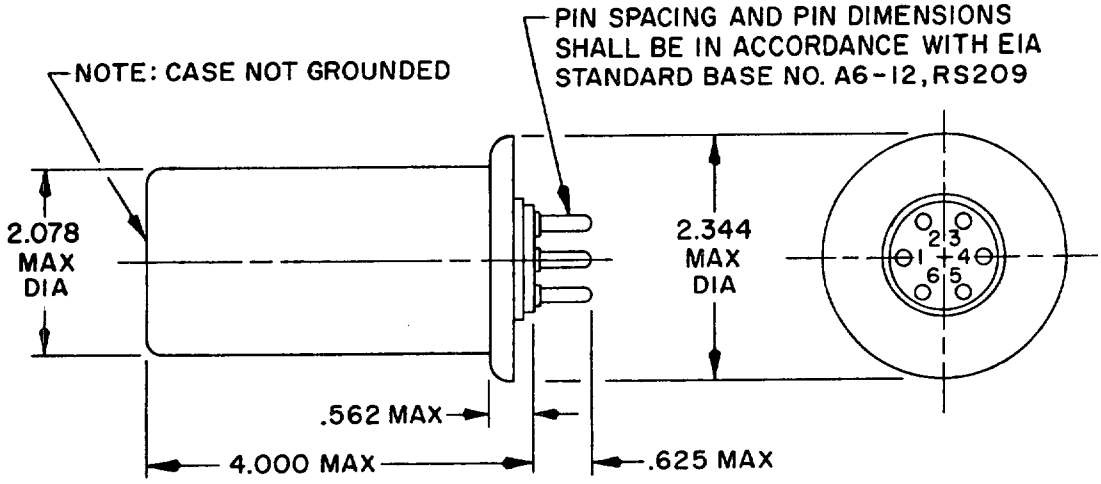
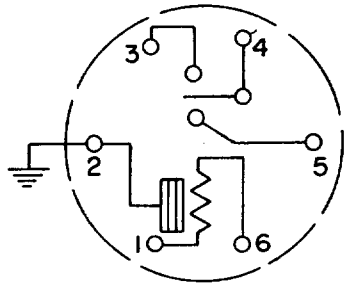
MILITARY SPECIFICATION SHEET

# CHOPPERS, ELECTROMECHANICAL, DUST TIGHT, SINGLE POLE, DOUBLE THROW, MAKE BEFORE BREAK

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the chopper described herein shall consist of this specification and the latest issue of MIL-C-83729.




SCHEMATIC DIAGRAM TO APPEAR ON SIDE OF CAN

| INCHES | MM |
|---|---|
| .562 | 14.27 |
| .625 | 15.88 |
| 2.078 | 52.78 |
| 2.344 | 59.54 |
| 4.000 | 101.60 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is ±.005 (.13 mm).
(B) 4. When coil is de-energized center reed may be in any of 3 positions.

FIGURE 1. Outline drawing, terminal arrangement and schematic diagram.
(B) denotes changes


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REQUIREMENTS:

Contacts:

Single pole, double throw, make before break.

Polarity: When pin 1 is positive, pin 4 is connected to pin 5 and disconnected from pin 3. When pin 6 is positive, pin 4 is connected to pin 3 and disconnected from pin 5.

Voltage and current: 1 microvolt to 6.0 volts, maximum. .001 ampere maximum.

Coil data:

Voltage rated: 6.3 volts maximum.

Resistance: 131 ohms ±10 percent.

Impedance: 132 ohms ±10 percent.

Frequency range: 50 to 65 Hz.

Waveform characteristics (at 25°C):

Dwell time: Each contact closed 198° ±7° of each cycle. Contacts closed simultaneously for 18° ±10 percent twice in each cycle.

Dissymmetry: Within ±7°.

Transfer time: Not applicable.

Phase angle: 17° ±5°.

Noise level: Stray pick up.

Electrostatic: $2 \times 10^{-10}$ V/ohm of input circuit impedance.

Electromagnetic: Less than $2 \times 10^{-6}$ volts constant to within $2 \times 10^{-7}$ volts.

Environmental conditions:

Dielectric withstanding voltage: Applicable.

Insulation resistance: Applicable.

Vibration II: Method 204, test condition A, MIL-STD-202.

Qualification and quality conformance inspection: 1 1/2 hours in each of three mutually perpendicular directions, including the most critical axis.

Output voltage: Vary less than 2 percent from 0 to 10Gs during vibration.

Shock: 30Gs for 11 milliseconds in three directions.

Temperature range: 25°C ±10°C.

Life:

Duration of test: 2,500 hours continuous.

Temperature: 25° ±2°C.

Coil voltage and frequency: Nominal.

Operating conditions: See table I.

Output voltage: Vary not more than 2 percent during life.

Contact resistance: Shall not exceed .100 ohm during and at conclusion of life.



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TABLE I. Life operating conditions.

| Contact voltage | Contact current | Hours |
|---|---|---|
| Volt | Ampere | |
| .010 | .001 | 1,000 |
| 1.0 | .0001 | 1,500 |

Mechanical:

Weight: .68 pound, maximum.

Dimensions: See figure 1.

Enclosure: Dust-tight case.

Terminals: Plug in, see figure 1.

Mounting poisition: Vertical for optimum performance.

Part number: M83729/2-001.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AT, AV, ER, MI, SL
Navy - AS, OS
Air Force - 11, 17
DLA - ES

User activities:
Army - ME, WC
Navy - MC
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Air Force - 85

(Project 5945-0643-2)

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